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Data Mining (CMPS452)

Heart Disease Prediction Using Random Forest Classifier

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Abstract

Heart disease is one of the top causes of death around the world, making early prediction important for reducing risks. This project aims at predicting whether a patient has any heart disease or not using machine learning.

Dataset we used here is heart disease dataset that contains several medical features like age, cholesterol, blood pressure, heart rate etc. We have mainly used Random Forest Algorithm as it can yield high accuracy and deal with multifeatured problems. Along with Random Forest we have also implemented Decision Tree Algorithm and compared which model performs better.

We have split the dataset into two parts Training and Testing Dataset to avoid biased performance. Our result shows that Random Forest outperforms Decision Tree in terms of accuracy.

This project is one examples of implementing data mining in healthcare sector for early diagnosis and understanding the impact of using right algorithm for prediction.

Chapter 1. Introduction

Main Problem Description

Heart disease is a major issue that affects people all around the world. It is just one of those diseases that kills a lot of people every year, and many of those deaths are caused by delayed detection. Traditional ways to diagnose it take forever, with all those tests and needing some specialist to look everything over. However, with the amount of medical data we have, it appears that data mining can make prediction better. For example, gathering data on age, cholesterol, or even heart rate and identifying trends that indicate the presence of the illness.

Motivation

This project applies machine learning techniques to predict early on if a person has heart disease. If doctors are informed earl on, they can take measures to avoid any possible complications. Using data sets and existing algorithms in machine learning, this project will apply them to predict a patient using certain attributes whether they are at risk for heart disease. This project will also identify the use of data mining techniques to solve real world problems, particularly in the health domain. It aims to utilize machine learning techniques which helps in decreasing human error and obtaining results quickly. This is crucial for medical diagnosis.

Project Objective

Predict Patient Heart Disease. The main goal of this project is to create a predictive model that can classify whether a patient has heart disease. We used the Random Forest algorithm as the primary model because of its high-performance rate and reliability. A decision tree model was also created and compared to the random forest model in order to assess the model's efficiency. Analyzing the dataset, understanding the features of each row, and getting a high accuracy with sound results were also crucial goals in this project.

Contribution of the Project

We apply machine learning to the medical dataset. First, we do the data preprocessing. We prepare the dataset and choose relevant features. After that we use two algorithms: Random Forest and Decision Tree. We train and evaluate models on the provided dataset by calculating accuracy, precision, and recall. In the end we compare the results of two models to see which one is better. This project demonstrates the significance of data mining in resolving problems of healthcare and medicine.

Chapter 2. Related Work

Algorithms and Applications

Predicting heart disease has been a widespread problem which many researchers have tried to solve with machine learning approaches. Decision Trees are a very common decision stump. They are simple to interpret because they just follow a tree structure, following different branches at different feature values. They have been applied to healthcare applications before, for instance to classify between patients and healthy individuals based on their symptoms and medical features. Decision Trees however have the tendency to overfit their training set, therefore performing poorly on new, unseen test cases.

Logistics Regression is another popular technique used for classification problems. It is commonly employed for binary classification problems such as disease diagnosis, for this example heart disease. The method is widely used and provides an added layer of understanding to the results. In many studies, a clean dataset with the right features, Logistic Regression has shown to perform well. However, it fails when the relationship between the features is not straightforward.

Support Vector Machines (SVM) another method that can be used for classification problems when trying to predict medical events. Like the Decision Trees and Random Forest methods mentioned above, SVM tries to optimize the boundary between the two classes to get the best separating hyperplane. SVM is a very powerful tool that has been widely used for classification tasks and shown to be very accurate in many domains, even the medical ones.

Predicting inpatient mortality has long been an important goal in intensive care units. Existing approaches to this problem include time series analysis, vector autoregression, and machine learning decision trees. Most recent approaches are based on ensemble decision trees, particularly Random Forest, and have been demonstrated to be effective in several domains, including disease prediction and patient stratification.

Related Algorithms

Random Forest is an ensemble learning method which uses numerous trees and averages (or otherwise combines) their results. Using Random Forests can improve performance and decrease overfitting when working with large datasets of medical images and data. It can be used as a classification or regression method.

Decision Tree algorithms split data and display the results in a tree format, using the feature values in the data. The Decision Tree is straightforward to understand and easy to visualize. While a Decision Tree algorithm will not necessarily yield the best accuracy, it is powerful for displaying results.

Logistic Regression corresponds to Binary Classification, where we use a logistic regression learning algorithm to train a classifier. Easy to understand and train, it is however less accurate than a Decision Tree as it assumes a linear relationship between the inputs and the outputs.

K-Nearest Neighbors (KNN) another common method like Decision Trees, assigns a class to unclassified data points based off the classes of their closest neighbors in the existing database of data. KNN is very simple to train but can be very slow for large data sets. It is also very prone to the effects of noisy data.

Most of the existing algorithms have been already used and implemented in the existing healthcare data mining projects and this assignment used Random Forest algorithm. This was chosen as it outperformed the other algorithms in terms of accuracy and handled the high dimensional feature space effectively.

Chapter 3. Dataset Description

Dataset Source

The dataset used is the well-known Heart Disease Dataset, that has been often used in literature for a variety of machine learning experiments. The dataset is composed of information for 608 patients who have been diagnosed for heart disease along with other relevant medical information. The data was collected and constructed to be used by training a model or predicting future data that would help identify individuals who have heart disease. The dataset includes 573 records (samples), each record includes 14 fields (features): 13 input attributes and 1 target output attribute. Many of the features include human interpretation such as age, sex, type of sugary and number of disease arteries. Other features require analytical interpretation in relation to heart health like total cholesterol, blood pressure and maximum heart rate.

Feature Description

Every column in the dataset is a feature which represents either a medical measurement or a patient attribute. For example, “age” represents the age of the patient and “sex” denotes whether the patient is male or female. The “cp” feature denotes they type of chest pain which is important indicator for heart disease. The “chol” feature represents the cholesterol level which indicate whether a patient has risk of heart disease or not. The “trestbps” and “thalach” features denote the resting blood pressure and maximum heart rate of a patient respectively. Features such as “exang” denote whether a patient has chest pain when they are exercising or not. The most important feature is “target” which denotes whether the patient has heart disease (1) or not (0).

Class Distribution

Our dataset includes patients with heart disease and patients without heart disease. Both classes are relatively balanced within the data and obtaining a balanced data set is key to developing a good model. A blanaced data set prevents any imbalance and makes it easier for a machine learning model to learn and accurately make predictions for both classes.

Chapter 4. Methodology

Type of Learning

For this project, we decided to use supervised learning because the data was provided, and it had labels. The data had many input features (age, sex, resting blood pressure, etc.) and it had an output as well. The output was several 1s and 0s indicating whether a person had heart disease. We used supervised learning because the data was labeled, and it allowed us to train a model to make predictions on new data. This specific problem was a classification problem because the output had two possible numbers, 0 or 1. Supervised learning is commonly used in healthcare to make accurate predictions to better treat patients using high quality data.

Proposed Algorithm

For this project, the main algorithm utilized was the Random Forest algorithm. Random Forest is an ensemble learning method that combines many individual decision trees, using a majority vote to average the output of the predictions of each decision tree. All the decision trees are trained with a subset of the features and a subset of samples. The Random Forest is a good choice for this task, because it can deal with many features, learns complex correlations between the features of a dataset and is robust to noise and missing values.

Why This Algorithm?

For the model selection, we have chosen to use Random Forest instead of single Decision Tree. This is because single decision trees suffer from overfitting problem. However, this problem can be significantly reduced by forming a forest of trees, Random Forest is capable of handling both numerical and categorical data. It also performs very well for medical datasets which typically contain large number of features. Data preprocessing is not required for this model, which makes it even easier to use. The model can predict heart disease with high accuracy and consistently. Other options like using Logistic Regression or SVM may have been good alternatives, but Random Forest is relatively easier to use more effective for most cases.

Algorithm Steps

Random Forest applies the Random Forest algorithm to a given dataset. First the dataset is loaded, then the input features and target are separated from the data. The data is then split into a training set and a test set, usually split 80/20. A Random Forest is many Decision Trees, each made from random subset of your training data. Each tree will predict something for the input features and the trees will vote for the majority answer. Then we can run the same algorithm, on our test data. The same process is run on the Decision Tree model for comparison. The results are then scored using accuracy, precision, and recall, and f1 score metrics.

System Workflow

This flow describes a process of preparing the dataset, training the Random Forest model and Decision Tree model on the training set, and then testing these models on the testing set. The performance of both models is evaluated by various performance metrics. The model that performs better is identified. This flow illustrates the entire process from the beginning to the end.

Data Preprocessing

To improve the performance of the model, data preprocessing is very important. Before applying any machine learning algorithm, we check for missing values in the dataset and then perform necessary cleaning (if missing values exist in the dataset used for this project, it was already cleaned up, hence minimal cleaning as performed in this case). Normalization/Scaling can also be performed for normalizing the range of values to improve model's performance (though it may not be required for Random Forest Algorithm). We could also perform feature selection to identify the most relevant features that affect the prediction and remove the features that are irrelevant to the target variable to improve model's performance. Finally, we would need to preprocess the data before feeding it to the machine learning algorithm for training.

Chapter 5. Experimental Setup

Train and Test Split

Data partitioning is an important step in a machine learning workflow. For this example, we will divide our data into two parts: a training set on which we train a model, and a test set to assess the quality of the learned model. Convention typically dictates an 80-20% split between training and testing data. Here, we are using such ratio and setting the random state so that in subsequent experiments, the same partitioning occurs.

Evaluation Metrics

To assess the quality of the models developed, standard performance measurements were employed including accuracy, precision, recall and F1 score. However. For a model intended for use in a medical setting such as heart disease prediction, high accuracy is not enough; there must also be good recall. In other words, the model should be able to remember to flag the cases of patients who have heart disease.

Tools and Libraries

This project is entirely written in Python. The basic library used is scikit-learn for both Random Forest and Decision Tree algorithms. The project uses the pandas library for the importing of the data. It uses NumPy library for numerical calculations. The code is uploaded in google colab for online execution. It can be simply copied and pasted online in the notebook to execute without downloading and installing library. The code is very simple and can be used to simply train a model, fit a model and to evaluate the performance of a model.

Chapter 6. Results and Comparison

Results of Proposed Model

After training the Random Forest model on our training dataset, we also applied it to our testing dataset. The model performed well on all evaluation metrics. This is great because it indicates that the model will have a high accuracy for classifying whether someone has heart disease based on the given dataset.

In addition to being accurate, the model is also precise. This means that most of the people the model predicts to have heart disease do have heart disease and should be concerned. Precision is often referred to as a health metric because you don't want to add to people's stress by telling them they have heart disease when they do not. Our deep learning solution has also a very high recall, meaning it is very good in identifying all people with heart disease and thus helping to save their lives.

In addition to the high accuracy and very good f1 score, the precision and recall also underline the good overall performance of the model. This used Random Forest model is a strong choice for this task and can learn from the dataset and predict well.

Results of the experiments are given in the table below, summarizing accuracy, precision, recall and F1 score for all the used models.

For this data, I used a Random Forest model to train, validate and test the data. It performed very well and can be used to predict heart disease.

Comparison with Related Algorithms

This project compares performance of Random Forest model and Decision Tree model. Both models were trained and evaluated on the same data using same train/test split and results generally show that Random Forest outperformed the Decision Tree model according to most evaluation metrics.

Using decision Tree model produced a lower accuracy rate when compared to the use of a Random Forest model. Decision Trees are prone

to overfitting, so they perform poorly when being used to make predictions on data outside of the training set. Random Forest, on the other hand, improves the performance of the model by reducing the overfitting through the creation of multiple trees in a forest.

Random Forest performed better than Decision Tree in terms of precision, recall and f1 score. It also had less error in terms of false positives. So, it performed better in terms of identifying actual heart disease cases as having heart disease.

The comparisons between the two models can be shown in the following table:

Model	Accuracy	Precision	Recall	F1-score
Random Forest	0.973	0.981	0.962	0.971
Decision Tree	0.956	0.980	0.925	0.952

The table shows that the Random Forest model outperformed the simple Decision Tree for this project. Results show that ensemble models are better than single models.

The results show that the performance of predicting heart disease of patients using Random Forest is better than other methods for this dataset.

Chapter 7. Discussion

In this project we have seen that Random Forest model outperforms Decision Tree approach for predicting heart disease. Random Forest model performed with higher accuracy, precision, recall and F1 score than Decision Tree. This is because Random Forest provides a combination of multiple Decision Trees and hence it does not over fit to the training data generalization is better.

We made the model and it seemed to accurately classify the patients. Age, cholesterol, and heart rate were all strong predictors. The mode learned to use these attributes to make predictions.

Decision Tree is one of the simple and most intuitive learning models, however it has very notable drawbacks: it tends to overfit training data. This means that its predictive power on new examples typically is low. In contrast, the performance of Random Forest model is much more stable and consistent (it generalizes well).

First and foremost: this is not a trivial task and we encountered couple of challenges during the process. The most significant of them was dataset size just because of its limited size we had serious doubts whether the system would perform well on much larger datasets. Additionally, not all the provided features might have the same importance; some of them are even irrelevant. Finally, performance of machine learning models is heavily dependent (if not determined) by the quality of training data.

Although there are a few issues to fix, we are generally happy with the results. There are certainly ways to improve the predictions, but overall, this project demonstrates the power of machine learning for medical prediction and shows how data mining can be used to help aid in the decision-making process and in early detection of many diseases.

Chapter 8. Conclusion

Our objective for this project was to develop a predictive model which we could determine of a patient has heart disease given certain medical data. To accomplish this, we utilized a Random Forest model for our primary predictions as well as a Decision Tree model for comparison. First, we analyzed the data, preprocessed it. Once we preprocessed the data, we split it into two components: a training set and test set.

From the results we obtained, it seems that using a Random Forest model produced the best accuracy, precision, recall and F1 scores for this task. Comparing to a simple Decision Tree allowed us to understand the benefits of ensemble learning as opposed to regular single learning models.

Healthcare data analytics is a very promising area, and this project explores the possibilities of Machine Learning techniques to support early diagnosis of diseases using large datasets of patients.

In addition to the classification task using different algorithms on the preprocessed data set, we also experienced the challenges of the project in the process of data preprocessing, model selection and evaluation.

Chapter 9. Future Work

There are several possible improvements to the project. For starters using a larger dataset could boost the accuracy of the model's predictions. Generally having more data means better learning for a model.

Alternatively, other machine learning techniques could be employed such as Support Vector Machines, Neural Networks or Gradient Boosting depending on the data structure.

In addition to improving the quality of the learned models, also the selected input data can be improved by selecting the most relevant features and removing redundant data.

Future development could easily involve further expanding upon the code to incorporate real time capabilities with a front end where user could load their data and have the model predict based on it.

There are many ways the project can be taken further to improve accuracy and apply it in real world settings.

Chapter 10. References

- **https://www.kaggle.com/datasets/anushamayaluri/cleveland-statlog**

- **https://uci-ics-mlr-prod.aws.uci.edu/**

- **https://www.stat.berkeley.edu/~breiman/randomforest2001.pdf**

- **https://www.semanticscholar.org/paper/Induction-of-Decision-Trees-Quinlan/bcee7c85d237b79491a773ef51e746bbbcf48e35**

- **https://colab.research.google.com/**

Chapter 11. Appendix

[1]
✓ 2s

```
# 1. Import Libraries
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestClassifier
from sklearn.tree import DecisionTreeClassifier
from sklearn.metrics import accuracy_score, precision_score, recall_score, f1_score
```

[2]
✓ 0s

```
# 2. First 5 rows in the DataSet

df = pd.read_csv("Dataset.csv")

print("First 5 rows:")
print(df.head())
```

```
First 5 rows:
   age  sex  cp  trestbps  chol  fbs  restecg  thalach  exang  oldpeak  slope  \
0   70    1   3       130   322    0        2      109      0      2.4      1
1   67    0   2       115   564    0        2      160      0      1.6      1
2   57    1   1       124   261    0        0      141      0      0.3      0
3   64    1   3       128   263    0        0      105      1      0.2      1
4   74    0   1       120   269    0        2      121      1      0.2      0

   ca  thal  target
0   3     1       1
1   0     3       0
2   0     3       1
3   1     3       0
4   1     1       0
```

[3] ✓ 0s

```
# 3. Prepare the Data

X = df.drop("target", axis=1)
y = df["target"]
```

[4] ✓ 0s

```
# 4. Split the data

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
```

[5] ✓ 0s

```
# 5. Train Random Forest

rf = RandomForestClassifier()
rf.fit(X_train, y_train)

rf_pred = rf.predict(X_test)
```

[6] ✓ 0s

```
# 6. Train Decision Tree

dt = DecisionTreeClassifier()
dt.fit(X_train, y_train)

dt_pred = dt.predict(X_test)
```

[7]
✓ 0s

```python
# 7. Evaluation

def evaluate_model(name, y_test, y_pred):
    print("\n", name)
    print("Accuracy:", accuracy_score(y_test, y_pred))
    print("Precision:", precision_score(y_test, y_pred))
    print("Recall:", recall_score(y_test, y_pred))
    print("F1-score:", f1_score(y_test, y_pred))
```

[8]
✓ 0s

```python
# 8. Results

evaluate_model("Random Forest", y_test, rf_pred)
evaluate_model("Decision Tree", y_test, dt_pred)
```

```
 Random Forest
Accuracy: 0.9652173913043478
Precision: 0.9807692307692307
Recall: 0.9444444444444444
F1-score: 0.9622641509433962

 Decision Tree
Accuracy: 0.9739130434782609
Precision: 0.9811320754716981
Recall: 0.9629629629629629
F1-score: 0.9719626168224299
```